

```
#1. Activate your toolkits
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 2. Load the built-in practice dataset
df = sns.load_dataset('tips')

# 3. Print a success message
print("Success! Your tools and data are ready.")
```

Success! Your tools and data are ready.

```
# 1. See the raw data (the first 5 rows of spreadsheet)
print("--- FIRST 5 ROWS ---")
display(df.head())

# 2. Check the data structure (Info)
print("\n--- DATA INFO ---")
df.info()

# 3. Get statistical summaries (Describe)
print("\n--- STATISTICAL SUMMARY ---")
display(df.describe())
```

--- FIRST 5 ROWS ---

	total_bill	tip	sex	smoker	day	time	size	
0	16.99	1.01	Female	No	Sun	Dinner	2	
1	10.34	1.66	Male	No	Sun	Dinner	3	
2	21.01	3.50	Male	No	Sun	Dinner	3	
3	23.68	3.31	Male	No	Sun	Dinner	2	
4	24.59	3.61	Female	No	Sun	Dinner	4	

--- DATA INFO ---

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 244 entries, 0 to 243
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   total_bill  244 non-null    float64
1   tip         244 non-null    float64
2   sex        244 non-null    category
3   smoker     244 non-null    category
4   day        244 non-null    category
5   time       244 non-null    category
6   size       244 non-null    int64
dtypes: category(4), float64(2), int64(1)
memory usage: 7.4 KB
```

--- STATISTICAL SUMMARY ---

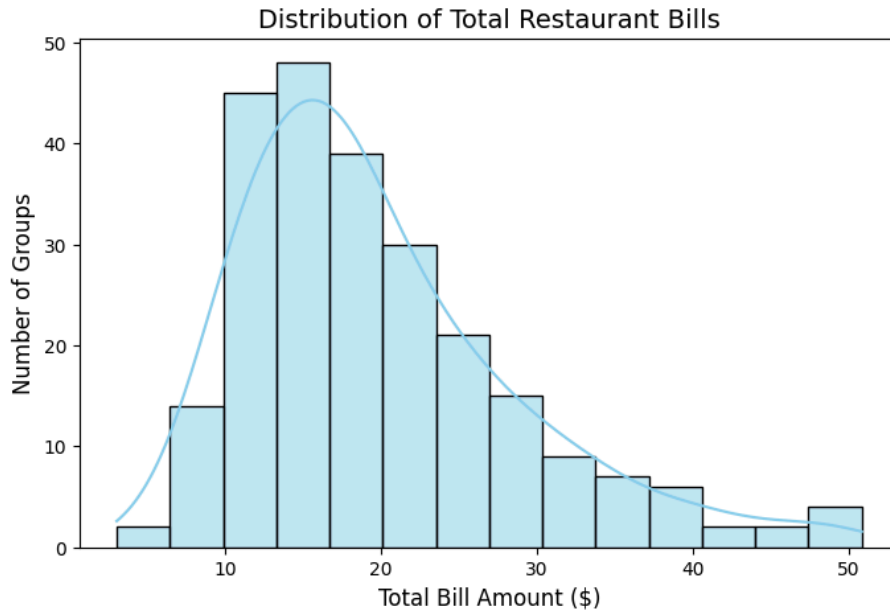
	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

```
#1. Tell the software how big to make chart
plt.figure(figsize=(8, 5))

# 2. Create the distribution plot (histogram) for total bills
sns.histplot(data=df, x='total_bill', kde=True, color='skyblue')
```

```
# 3. Add titles so it looks like a professional report
plt.title('Distribution of Total Restaurant Bills', fontsize=14)
plt.xlabel('Total Bill Amount ($)', fontsize=12)
plt.ylabel('Number of Groups', fontsize=12)

# 4. Display the chart
plt.show()
```



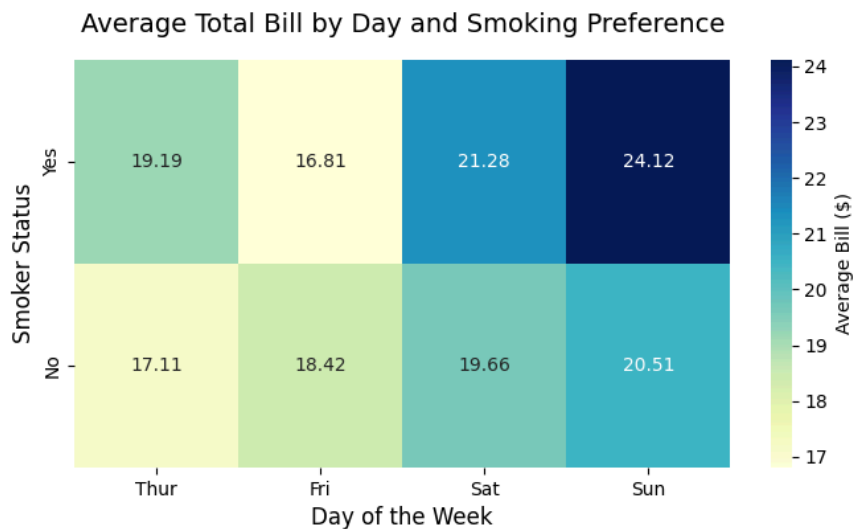
```
# 1. Arrange the data into a grid format (Days vs Smoker status)
grid_data = df.groupby(['smoker', 'day'], observed=False)['total_bill'].mean().unstack()

# 2. Tell the software how big to make chart
plt.figure(figsize=(8, 4))

# 3. Create the heatmap with colors and number labels
sns.heatmap(grid_data, annot=True, fmt=".2f", cmap="YlGnBu", cbar_kws={'label': 'Average Bill ($)'})

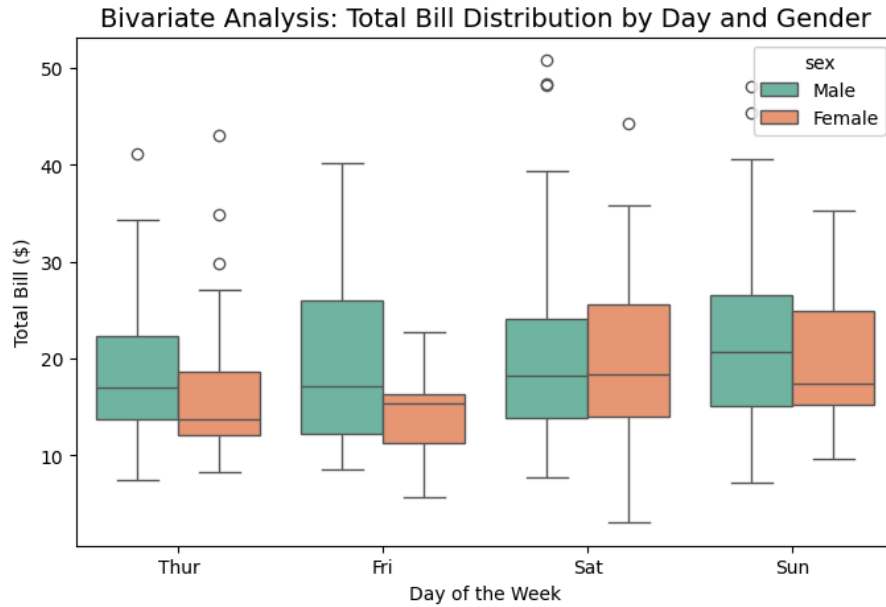
# 4. Add titles to make it look professional
plt.title('Average Total Bill by Day and Smoking Preference', fontsize=14, pad=15)
plt.xlabel('Day of the Week', fontsize=12)
plt.ylabel('Smoker Status', fontsize=12)

# 5. Display the chart
plt.show()
```

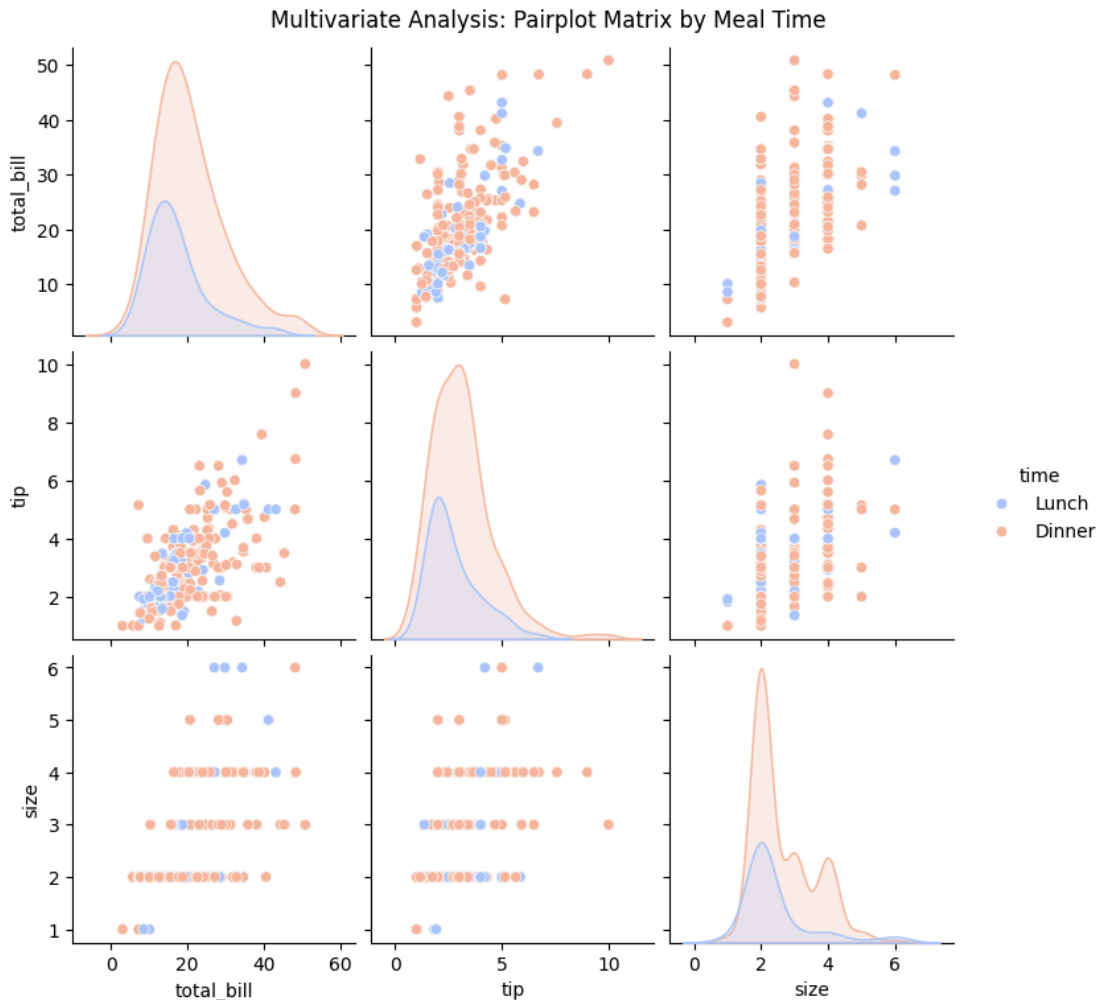


```
#Bivariate Analysis (Comparing Total Bill across Days of the week)
plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x='day', y='total_bill', hue='sex', palette='Set2')
```

```
plt.title('Bivariate Analysis: Total Bill Distribution by Day and Gender', fontsize=14)
plt.xlabel('Day of the Week')
plt.ylabel('Total Bill ($)')
plt.show()
```



```
#Multivariate Pairplot to inspect correlations and check for multicollinearity
sns.pairplot(df, hue='time', palette='coolwarm')
plt.suptitle('Multivariate Analysis: Pairplot Matrix by Meal Time', y=1.02)
plt.show()
```



Task 5: Exploratory Data Analysis (EDA) Report

1. Data Structure Observations (Hint A)

- **Dataset Scope:** The dataset consists of 244 individual restaurant transaction records.
- **Key Metrics:** The average total bill across all dining groups is approximately 19.78, *with a standard deviation of 8.90*.

2. Visual Analysis Insights (Hints B, C, D, & E)

- **Distribution Analysis (Hint B):** The total bill distribution is heavily right-skewed. The majority of dining groups spend within the 15 to 20 range, with a few high-spending outliers pulling the tail of the graph outward.
- **Bivariate Relationships (Hint C):** When comparing spending behavior across days, total bills increase significantly during the weekend (Saturday and Sunday) compared to weekday dining (Thursday and Friday).
- **Matrix Heatmap Patterns (Hint D):** Looking at the breakdown of days and smoking preference, the highest average spending occurs on Sundays among smoking parties (24.12), *while the lowest occurs on Thursdays among non-smokers* (17.11).
- **Multivariate Correlations (Hint E & F):** The overall pairplot reveals a strong, positive linear correlation between the `total_bill` and the `tip` amount—meaning as the bill increases, the tip increases proportionally.